VIF^2 and FastFood: Visual-Ingredient Feature Fusion and a Large-Scale Benchmark for Food Nutrition Estimation

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ABSTRACT

Estimating calories and macronutrients from a single food image remains difficult because large-scale nutrition labels are scarce, food appearance is weakly tied to underlying composition, and many existing systems depend on portion or depth cues that are not always available. We present two complementary contributions: FastFood, a new benchmark for food nutrition estimation, and VIF^2, a visual-ingredient feature fusion method that explicitly injects ingredient semantics into image representations. FastFood contains 84,446 images covering 908 branded fast-food menu items from 10 global brands, each annotated with calories, fat, carbohydrates, protein, and a standardized ingredient list. The dataset is built from official nutrition sources and expanded with web retrieval to increase appearance diversity while preserving reliable labels. VIF^2 encodes ingredient text with a frozen CLIP text encoder, projects the resulting representation into the visual feature space, and fuses it with CNN or ViT backbones for multi-task nutrition regression. On FastFood, VIF^2 reduces ResNet101 caloric MAE from 118.04 to 61.26 and consistently improves fat, carbohydrate, and protein estimation. The approach also transfers to Nutrition5k, where ResNet101 improves from 103.03 to 83.75 caloric MAE under direct prediction and reaches 23.32 caloric MAE in a mass-aware setting. Additional analyses show that intermediate-layer fusion is strongest, LLaVA1.6 provides the best test-time ingredient generation, and majority voting over augmented views further lowers error.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision tasks**; • **Applied computing** → *Health care information systems.*

KEYWORDS

food nutrition estimation, ingredient-aware learning, multimodal fusion, benchmark dataset, dietary analysis

ACM Reference Format:
Anonymous Authors. 2025. VIF^2 and FastFood: Visual-Ingredient Feature Fusion and a Large-Scale Benchmark for Food Nutrition Estimation. In *Proceedings of the 2025 International Conference on Multimedia Retrieval (ICMR '25), June 30-July 3, 2025, Chicago, IL, USA.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3731715.3733269

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ICMR '25, June 30-July 3, 2025, Chicago, IL, USA
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ACM ISBN 979-8-4007-1877-9/2025/06
https://doi.org/10.1145/3731715.3733269

1 INTRODUCTION

Nutrition estimation from food images is a core problem in food computing because it enables dietary logging, health monitoring, and nutrition-aware retrieval without requiring manual entry [9]. Despite steady progress in food recognition benchmarks such as Food-101, Recipe1M, and Food2K [1, 10, 12], accurate prediction of calories and macronutrients remains challenging. Visual appearance alone is often insufficient: two dishes with similar presentation can differ substantially in ingredients and nutrition, while two images of the same branded product can vary in lighting, viewpoint, packaging, and plating. Prior nutrition-estimation systems have therefore leaned on depth, portion, or multi-view cues [6, 14, 16], but such signals are not always available in real deployment.

Ingredients provide a complementary source of information. They are much more directly related to nutrition than raw pixels, and the food literature has repeatedly shown that ingredient reasoning is valuable for recognition, retrieval, and recipe understanding [2, 3, 5]. However, ingredient supervision is rarely integrated into nutrition estimation benchmarks at scale. Existing datasets either provide accurate nutrition but limited scale and controlled capture settings, as in Nutrition5k [16], or provide large-scale food imagery without precise paired nutrition labels [10, 12]. This leaves a gap between ingredient-aware food understanding and practical nutrition regression.

This paper addresses that gap with two contributions. First, we introduce FastFood, a large-scale benchmark of 84,446 images covering 908 fast-food menu items from 10 global brands. Each item is paired with standardized labels for calories, fat, carbohydrates, protein, and ingredients. Unlike tray-based capture setups, FastFood emphasizes real-world appearance diversity while retaining precise nutrition from official brand sources. Second, we propose VIF^2 (Visual-Ingredient Feature Fusion), a simple and model-agnostic approach that encodes ingredient text with CLIP [11] and injects the resulting representation into intermediate visual features. The method works with both convolutional and transformer backbones and supports test-time ingredient generation with large multimodal models (LMMs).

Our empirical results show that the combination matters. On FastFood, VIF^2 nearly halves ResNet101 caloric MAE, from 118.04 to 61.26, and delivers similarly large gains on fat, carbohydrate, and protein. On Nutrition5k [16], the same fusion strategy improves direct monocular prediction and also performs strongly in a mass-aware protocol. We further analyze fusion depth, backbone choice,

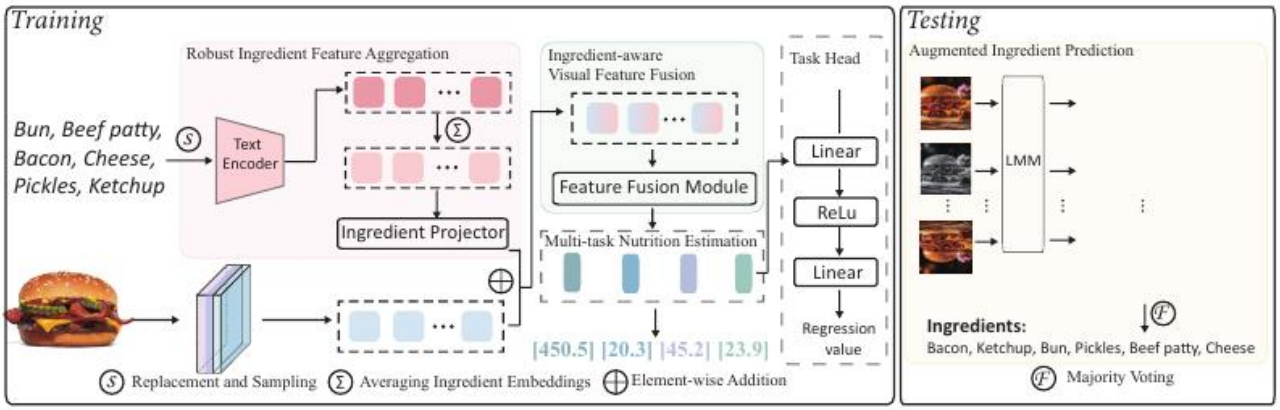

Figure 1: Overview of VIF^2. During training, ingredient strings are perturbed through replacement and sampling, encoded by a frozen CLIP text encoder [11], projected to the backbone feature dimension, and fused with visual features for multi-task prediction of calories, fat, carbohydrates, and protein. During testing, ingredients can be generated by a large multimodal model from multiple augmented views and consolidated with majority voting before fusion.

ingredient-generation models, and majority-vote thresholds, showing that the benchmark is diagnostic rather than merely larger.

Our main contributions are:

- We introduce FastFood, a large-scale nutrition benchmark with 84,446 images, 908 menu items, 10 brands, paired ingredient lists, and precise labels for calories and macronutrients.
- We propose VIF^2, a model-agnostic visual-ingredient fusion method that injects CLIP-encoded ingredient semantics into CNN and ViT backbones for multi-task nutrition regression.
- We present a test-time ingredient inference pipeline based on LMM predictions from augmented views and majority voting, and quantify its effect on downstream nutrition estimation.
- We provide a comprehensive benchmark study on FastFood and Nutrition5k, including backbone comparisons, layer ablations, LMM comparisons, threshold analysis, and qualitative error analysis.

2 RELATED WORK

2.1 Food Benchmarks and Nutrition Estimation

Food computing spans recognition, retrieval, recipe understanding, and dietary assessment [9]. Large-scale datasets such as Food-101 [1], Recipe1M [12], and Food2K [10] have accelerated progress in food recognition and cross-modal learning, but they do not provide the combination of precise per-item nutrition and standardized ingredient annotations needed for nutrition regression. Ingredient-centric datasets and studies, including Vireo Food-251 and related ingredient-recognition work [2, 3], show the value of ingredient supervision, but they do not directly target nutrition prediction with trusted nutrient labels.

Nutrition5k [16] is the most relevant public benchmark for nutrition estimation, offering high-quality annotations, depth data, and mass measurements for tray-captured meals. Subsequent methods have improved performance by predicting depth or fusing RGB with depth information [6, 13, 14]. These settings are important, but they emphasize controlled capture and portion reasoning rather than broad appearance variability and ingredient semantics. Our goal is complementary: we introduce a benchmark with substantially larger visual diversity and standardized branded nutrition, and we study ingredient-aware monocular estimation.

2.2 Ingredient-Aware Food Understanding

Ingredient prediction is a long-standing direction in food understanding because ingredients are more semantically stable than dish names and more informative for nutrition [2, 3]. Prior work has studied ingredient recognition, zero-shot ingredient prediction, and ingredient-aware recipe generation [2, 3]. More recently, vision-language models and LMMs have enabled richer food reasoning, including FoodLMM [18] and recipe-generation systems that predict ingredients before generating instructions.

Our method differs from prior ingredient-recognition work in two ways. First, we do not treat ingredient prediction as the final task; ingredients are used as a structured semantic signal for downstream nutrition regression. Second, we use frozen CLIP text features [11] and inject them directly into visual backbones, yielding a simple fusion mechanism that transfers across ResNet [7], InceptionV3 [15], and ViT [4]. At test time, we further exploit LMMs such as LLaVA1.6 [8] and Qwen2-VL [17] as ingredient generators rather than end-to-end nutrition estimators.

3 FASTFOOD BENCHMARK CONSTRUCTION

3.1 Scope and Design Goals

FastFood is designed to study monocular nutrition estimation under three constraints that are common in practice: diverse web imagery, reliable nutrition labels, and ingredient-aware reasoning. The dataset contains 84,446 images covering 908 branded menu items from 10 global fast-food brands. Each item is associated with

calories, fat, carbohydrates, protein, and a standardized ingredient list. We use a fixed 70/20/10 train/validation/test split at the image level.

We intentionally focus on branded fast food because it provides a rare combination of exact nutrition values from official sources and broad appearance diversity on the web. This makes the benchmark complementary to Nutrition5k [16]: Nutrition5k captures real prepared meals with depth and mass labels, whereas FastFood isolates the challenge of inferring nutrition from appearance and ingredient semantics in the wild.

3.2 Data Collection and Expansion

We begin from official brand websites, which provide canonical item names, product images, and nutrition values. These official pages serve as the source of record for labels. To expand visual diversity beyond studio product photos, we retrieve additional images from the web using two complementary strategies: image-based retrieval for visually clean item matches and text-based retrieval for broader appearance variation. Figure 3 illustrates the difference. Image-based retrieval tends to preserve the canonical appearance of an item, while text-based retrieval introduces more realistic variation in viewpoint, context, presentation, and imaging conditions.

We manually filter the retrieved images to remove mismatched products, wrong item sizes, severely blurred examples, and scenes containing multiple unrelated foods. Combo meals are retained only when they correspond to a specific labeled menu item. We also standardize item naming and units across brands to ensure consistent matching between images and nutrition tables.

3.3 Ingredient Annotation and Quality Control

Official nutrition pages do not always expose clean, model-ready ingredient lists, so we build a standardized ingredient annotation pipeline. For each menu item, a large multimodal model first proposes an ingredient list conditioned on the item name and reference image. Human annotators then refine the output to enforce specificity, remove irrelevant packaging or promotional phrases, merge duplicates, and standardize vocabulary across brands. This process yields ingredient strings that are suitable both for training and for later LMM-based prediction at test time.

Quality control is applied at both the image and annotation levels. Images are manually verified for menu-item consistency. Ingredient lists are checked for missing core components and vocabulary consistency. In addition to the nutrition-estimation benchmark, the dataset contains a supplementary dialogue resource of 5,800 images paired with 20,204 multi-turn diet dialogues, intended as a future resource for food-oriented instruction tuning. This dialogue subset is not used in the nutrition-regression experiments reported here.

3.4 Auxiliary Dialogue Resource

Beyond regression, FastFood includes 5,800 images with 20,204 multi-turn diet dialogues derived from the same nutrition records. Figures 5 and 8 document this auxiliary resource for evaluating multimodal dietary assistance.

3.5 Benchmark Tasks and Statistics

The primary benchmark task is multi-task regression of calories, fat, carbohydrates, and protein from a single RGB image. We report mean absolute error (MAE) and relative error, where the latter is the total absolute error normalized by the total ground-truth quantity over the evaluation split.

Figure 4 summarizes the benchmark distribution. The dataset spans a wide nutritional range, from low-calorie drinks and desserts to high-calorie sandwiches, burritos, and combo items. The ingredient distribution is also long-tailed: common ingredients such as bun, cheese, chicken, lettuce, and sauce appear frequently, while brand-specific or preparation-specific ingredients appear much less often. These properties make the benchmark useful for diagnosing when appearance is sufficient and when explicit ingredient reasoning is needed. Because labels are standardized at the menu-item level while imagery is gathered from diverse web sources, the benchmark also decouples nutritional identity from superficial presentation. Models therefore need robustness to packaging, plating, background clutter, and brand-specific photography rather than simple memorization of canonical product shots.

4 METHOD

4.1 Problem Setup

Given an input food image x, the goal is to predict a vector of nutrition targets $y = [y^{\text{cal}}, y^{\text{fat}}, y^{\text{carb}}, y^{\text{protein}}]$. During training on FastFood, we assume access to a standardized ingredient list $s = \{w_1, \ldots, w_n\}$ for each image. During testing, the ingredient list may come either from ground truth, for an oracle upper bound, or from an external LMM.

4.2 Ingredient Encoding

We encode each ingredient word or phrase with the frozen CLIP text encoder [11]. Let e_i denote the CLIP embedding of ingredient w_i. We aggregate the list by simple averaging:

$$e_{\text{ing}} = \frac{1}{n} \sum_{i=1}^{n} e_i.$$

The resulting ingredient embedding is projected to the feature dimension of the target backbone through a learned linear layer followed by ReLU. This projected vector is the ingredient signal injected into the visual network.

4.3 Robust Ingredient Feature Aggregation

Ingredient lists predicted at test time are noisy, so we make the training signal robust. In the FastFood training pipeline, we apply two perturbations to ingredient strings before CLIP encoding: synonym replacement and random ingredient-list subsampling. Specifically, ingredient names with known synonyms are replaced with probability 0.4, and with probability 0.5 we retain a random non-empty subset of the ingredient list. These perturbations force the model to rely on stable ingredient semantics rather than a brittle exact string match.

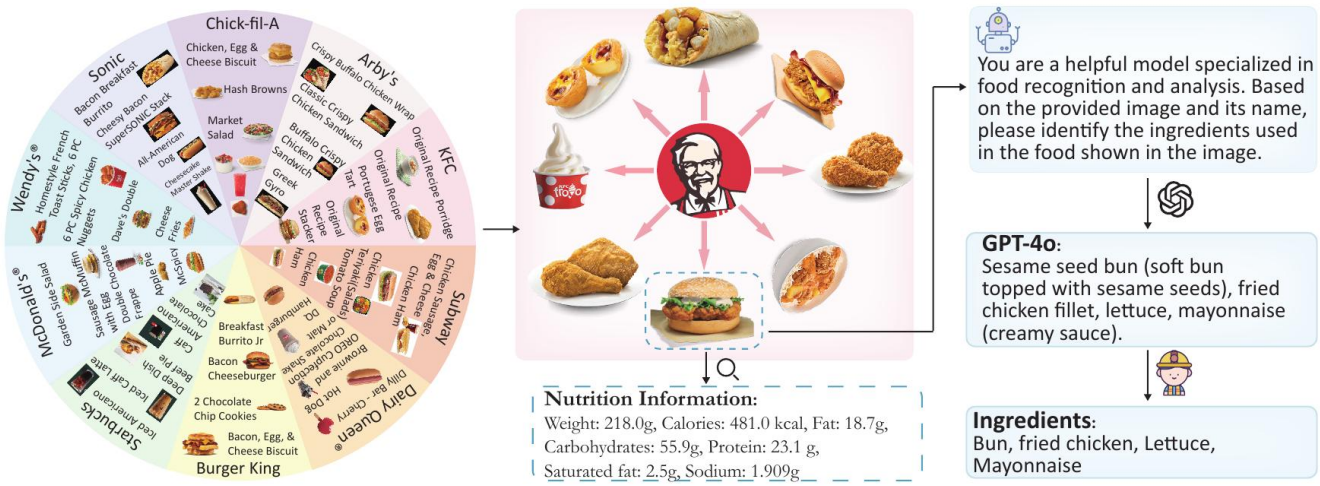

Figure 2: Overview of FastFood. The dataset spans 10 brands and 908 menu items, and pairs visually diverse images with standardized nutrition and ingredient annotations. Ingredient lists are initialized with a multimodal model and then manually refined for specificity and consistency.

Figure 3: Data expansion through image-based and text-based retrieval. The two retrieval modes are complementary: one preserves clean product identity, while the other adds harder in-the-wild appearance variation.

4.4 Visual-Ingredient Feature Fusion

Figure 1 shows the overall architecture. VIF^2 is designed to be backbone-agnostic.

CNN backbones. For ResNet and InceptionV3, the projected ingredient vector is broadcast spatially and added to an intermediate feature map. If the visual activation has shape $C \times H \times W$, the projected ingredient vector is transformed to dimension C and added channel-wise at each spatial location. This preserves the spatial structure of the image features while biasing them toward ingredient-consistent representations.

ViT backbone. For ViT [4], the projected ingredient feature is inserted as an additional token alongside the class token and patch tokens. The transformer then reasons jointly over image patches and the ingredient token.

Prediction heads. On top of the fused representation, we use task-specific MLP heads for calories, fat, carbohydrates, and protein. All targets are optimized jointly with a summed regression loss. In InceptionV3, we optionally include auxiliary heads during training. For the mass-aware Nutrition5k protocol, we additionally predict dish mass and nutrient density, then reconstruct total nutrient predictions by multiplying mass with the per-unit-mass outputs at test time.

4.5 Test-Time Ingredient Generation

To remove the requirement for ground-truth ingredients at test time, we use an external LMM to predict ingredients from the image. Because single-view predictions can be unstable, we query the LMM on multiple augmented views of the same image and retain ingredients that appear at least τ times across predictions. We refer to this as majority voting. The consolidated ingredient list is then passed through the same CLIP encoder and fusion module used during training.

5 EXPERIMENTS

5.1 Experimental Setup

We evaluate ResNet50, ResNet101, InceptionV3, and ViT backbones [4, 7, 15]. The CLIP text encoder is frozen [11]. Training uses standard image augmentation, including resize, random rotation, horizontal flip, and color jitter. We report MAE and relative error for all four nutrition targets.

We evaluate on two datasets. For FastFood, models are trained and validated on the 70/20/10 split described above. At test time, the default setting uses generated ingredients rather than ground truth; oracle results with ground-truth ingredients are reported separately as an upper bound. For Nutrition5k [16], we report two settings: Protocol 1 directly predicts total calories and macronutrients, while

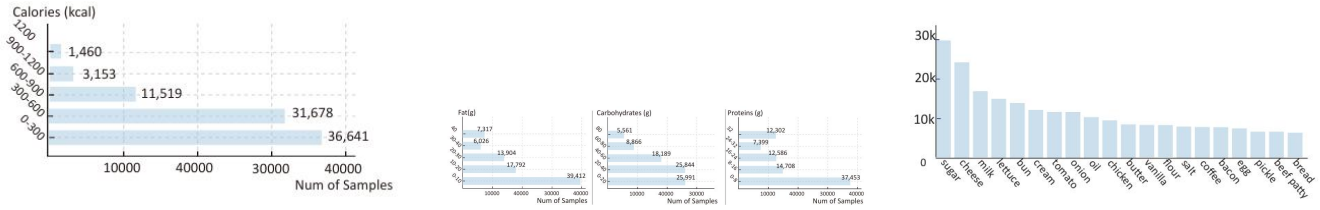

Figure 4: Benchmark statistics for FastFood. Left: calorie distribution. Middle: distributions over fat, carbohydrates, and protein. Right: frequency of the top-20 ingredients. Together they show broad nutritional coverage and a long-tailed ingredient space.

...I will provide a fast food image and its nutrion informaon. Please help me generate a multi-round dialogue about the image on dietary planning and suggestions as the model's training data.
For example, nutrition information description like this: TOTAL(194, 15.12, 0.5, 13.3) means this food item has 194 kcal calories and 15.12g fat, 0.5g carbohydrate and 13.3 protein.
One 1-round example is: \Image Nutrition Information: TOTAL (500, 23, 49, 24) **USER:** Can you provide some dietary suggestions based on this image? **ASSISTANT:** Sure! To balance this meal, consider pairing it with low-calorie, nutrient-rich side such as a green salad...
Nutrition Information: TOTAL ({cal}, {fat}, {carb}, {pro}). Please help me generate a {N}-round dialogue closely related to dietary planning and suggestions. The questions are expected to be professional or general, the answers are expected to be explanatory.

Figure 5: Prompt template for the auxiliary diet-dialogue subset.

Protocol 2 is a mass-aware setting that predicts mass together with nutrient density and converts them to absolute nutrient values.

5.2 Main Results on FastFood

Table 1 reports the main benchmark results. VIF^2 improves every backbone by a large margin, with the strongest results obtained by ResNet101. Caloric MAE drops from 118.04 to 61.26, a 48.1% reduction. The same trend holds for fat (7.06 to 3.60), carbohydrates (15.57 to 8.76), and protein (7.18 to 3.87). The gains are consistent across ResNet50, InceptionV3, and ViT, showing that the improvement is not tied to a single architecture.

5.3 Transfer to Nutrition5k

Table 2 shows that the ingredient-fusion strategy transfers beyond FastFood. Under Protocol 1, VIF^2 improves caloric MAE for all backbones, reducing ResNet101 from 103.03 to 83.75. It also sharply improves macronutrient estimation, with the best fat MAE at 6.94, best carbohydrate MAE at 5.58, and best protein MAE at 3.15. Under the mass-aware Protocol 2, ResNet101 + VIF^2 reaches 23.32 caloric MAE, 1.93 fat MAE, 1.57 carbohydrate MAE, and 0.91 protein MAE.

Table 1: Performance on FastFood using generated ingredients at test time. The best result in each column is boldfaced.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
ResNet101 [7]	118.04 / 29.84%	7.06 / 43.98%	15.57 / 39.14%	7.18 / 40.30%
ResNet50 [7]	118.07 / 29.85%	6.97 / 43.40%	15.32 / 38.52%	7.03 / 39.43%
ViT [4]	141.39 / 35.76%	8.20 / 51.07%	18.76 / 47.17%	8.58 / 48.17%
InceptionV3 [15]	142.57 / 36.04%	10.51 / 65.49%	22.74 / 57.18%	11.62 / 65.17%
ResNet101 + VIF^2	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
ResNet50 + VIF^2	63.86 / 16.15%	3.99 / 24.85%	9.67 / 24.32%	4.18 / 23.44%
ViT + VIF^2	76.62 / 19.38%	4.75 / 29.60%	12.01 / 30.20%	5.29 / 29.71%
InceptionV3 + VIF^2	80.86 / 20.44%	4.93 / 30.70%	11.02 / 27.72%	4.94 / 27.71%

Table 2: Performance on Nutrition5k. Protocol 1 directly predicts total nutrients. Protocol 2 is the mass-aware setting. The best result in each column within each protocol is boldfaced.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
		Protocol 1		
ResNet101 [7]	103.03 / 42.81%	9.94 / 69.35%	12.91 / 72.86%	7.56 / 56.56%
ResNet50 [7]	104.68 / 43.49%	9.97 / 69.58%	12.92 / 72.91%	7.64 / 57.09%
ViT [4]	148.47 / 61.69%	11.57 / 80.78%	12.67 / 71.48%	9.49 / 70.98%
InceptionV3 [15]	106.50 / 44.25%	9.93 / 69.33%	13.06 / 73.70%	7.72 / 57.73%
ResNet101 + VIF^2	**83.75 / 34.80%**	**6.94 / 48.45%**	5.64 / 31.85%	3.27 / 24.46%
ResNet50 + VIF^2	84.49 / 35.11%	7.01 / 48.95%	5.63 / 31.77%	**3.15 / 23.55%**
ViT + VIF^2	87.31 / 36.28%	7.26 / 50.71%	**5.58 / 31.49%**	3.24 / 24.19%
InceptionV3 + VIF^2	90.65 / 37.67%	7.63 / 53.27%	6.57 / 37.05%	3.75 / 28.02%
		Protocol 2		
ResNet101 + VIF^2	**23.32 / 9.69%**	**1.93 / 13.50%**	**1.57 / 8.87%**	**0.91 / 6.82%**
ResNet50 + VIF^2	28.84 / 11.98%	2.39 / 16.71%	1.92 / 10.84%	1.08 / 8.04%
ViT + VIF^2	31.07 / 12.91%	2.58 / 18.05%	1.99 / 11.21%	1.15 / 8.61%
InceptionV3 + VIF^2	31.81 / 13.22%	2.68 / 18.69%	2.31 / 13.00%	1.32 / 9.83%

This transfer is notable because Nutrition5k differs substantially from FastFood in capture protocol, meal composition, and the role of portion cues. The gains in both direct and mass-aware settings indicate that ingredient fusion is not merely exploiting branded menu regularities; it also improves estimation for mixed dishes where composition must be inferred under stronger portion uncertainty.

6 ANALYSIS

6.1 Where to Fuse Ingredient Features

Table 3 studies the fusion location. Intermediate fusion works best for both major CNN families. For ResNet101, injecting ingredients at block 2 yields the lowest average MAE, 20.47. For InceptionV3, the best stage is Mixed6e with the auxiliary head enabled, at 25.44. Very

Table 3: Ablation on fusion location. Average MAE is averaged over calories, fat, carbohydrates, and protein. The best result for each backbone is boldfaced.

Model	Injection layer	Avg. MAE	Model	Injection layer	Avg. MAE
ResNet101	block1	22.20 / 24.76%	InceptionV3	PostMaxPool2	27.10 / 28.26%
	block2	**20.47 / 22.22%**		Mixed6e (w/ aux)	**25.44 / 26.64%**
	block3	21.57 / 22.20%		Mixed6e (w/o aux)	25.66 / 26.77%
	block4	33.09 / 32.94%		PostMixed7c	32.75 / 33.55%

Table 4: Comparison of large multimodal models for test-time ingredient generation on FastFood. The best result in each column is boldfaced.

LMM	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
LLaVA1.6 (7B) [8]	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Qwen2-VL (7B) [17]	69.56 / 17.59%	4.30 / 26.82%	10.14 / 25.49%	4.54 / 25.45%
LLaVA-NeXT (Llama-3 8B) [19]	73.10 / 18.48%	4.50 / 28.03%	10.54 / 26.49%	4.63 / 26.00%

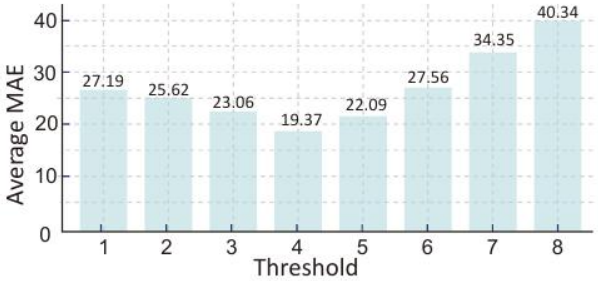

Figure 6: Average MAE over the four nutrition targets as a function of the majority-vote threshold τ for LLaVA1.6-generated ingredients. The best setting is $\tau = 4$.

late fusion is clearly worse, suggesting that ingredient information is most useful when it can influence mid-level semantic feature formation rather than only the final pooled representation.

6.2 Quality of Test-Time Ingredient Generation

Table 4 compares LMMs used to generate ingredients at test time. LLaVA1.6 is the strongest option among the tested models, producing the best MAE for all four nutrition targets. This result suggests that the downstream regressor is sensitive to ingredient fidelity and that open-ended multimodal generation is already strong enough to provide a useful test-time semantic prior.

Figure 6 studies the majority-vote threshold τ for LLaVA1.6-generated ingredients. The lowest average MAE occurs at $\tau = 4$, after which performance degrades as the retained ingredient set becomes too sparse. This confirms that modest consensus filtering is helpful, but aggressive filtering removes nutritionally important ingredients.

6.3 Upper Bound with Ground-Truth Ingredients

Table 5 reports results when ground-truth ingredients are provided at test time. The best model, ResNet101 + VIF^2, improves from 61.26 to 44.71 caloric MAE, indicating substantial remaining headroom in the ingredient-generation step. This gap is important: the fusion mechanism itself is stronger than the end-to-end pipeline suggests, and future gains can come from better ingredient acquisition without changing the regressor.

Table 5: Performance on FastFood when ground-truth ingredients are available at test time. This serves as an oracle upper bound for the ingredient-generation pipeline. The best result in each column is boldfaced.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
ResNet101 + VIF^2	**44.71 / 11.30%**	**2.59 / 16.13%**	**7.06 / 17.75%**	**2.85 / 15.99%**
ResNet50 + VIF^2	47.39 / 11.98%	3.08 / 19.21%	8.07 / 20.28%	3.21 / 18.02%
ViT + VIF^2	48.29 / 12.21%	3.42 / 21.29%	9.19 / 23.09%	3.41 / 19.12%
InceptionV3 + VIF^2	48.53 / 12.27%	3.07 / 19.15%	7.77 / 19.53%	3.06 / 17.18%

Figure 7: Qualitative examples with low and high prediction error. Distinctive, ingredient-sparse items are easier, while visually ambiguous or highly variable items remain challenging.

6.4 Qualitative Error Analysis

Figure 7 shows typical low-error and high-error cases. Items with distinctive visual structure and a compact ingredient list, such as breakfast sandwiches and beverages, are predicted accurately. Large errors tend to occur for visually ambiguous items with substantial intra-class variation, heavy occlusion, or poorly localized nutrient-dense components. The mac-and-cheese tray example is representative: texture alone provides weak cues about portion thickness and preparation details, and errors in ingredient prediction can further propagate into nutrition regression.

Overall, the analysis suggests three takeaways. First, the benchmark is genuinely diagnostic: it reveals strong dependence on ingredient semantics even for modern visual backbones. Second, mid-level fusion is the most effective operating point. Third, the current bottleneck is increasingly test-time ingredient quality rather than the fusion model itself.

7 CONCLUSION

We presented FastFood, a large-scale benchmark for food nutrition estimation, and VIF^2, a simple visual-ingredient feature fusion framework for multi-task prediction of calories and macronutrients. The benchmark fills a gap between large-scale food imagery and precise nutrition supervision by pairing diverse fast-food images with standardized ingredient and nutrition labels. The method improves every evaluated backbone on FastFood and transfers well to

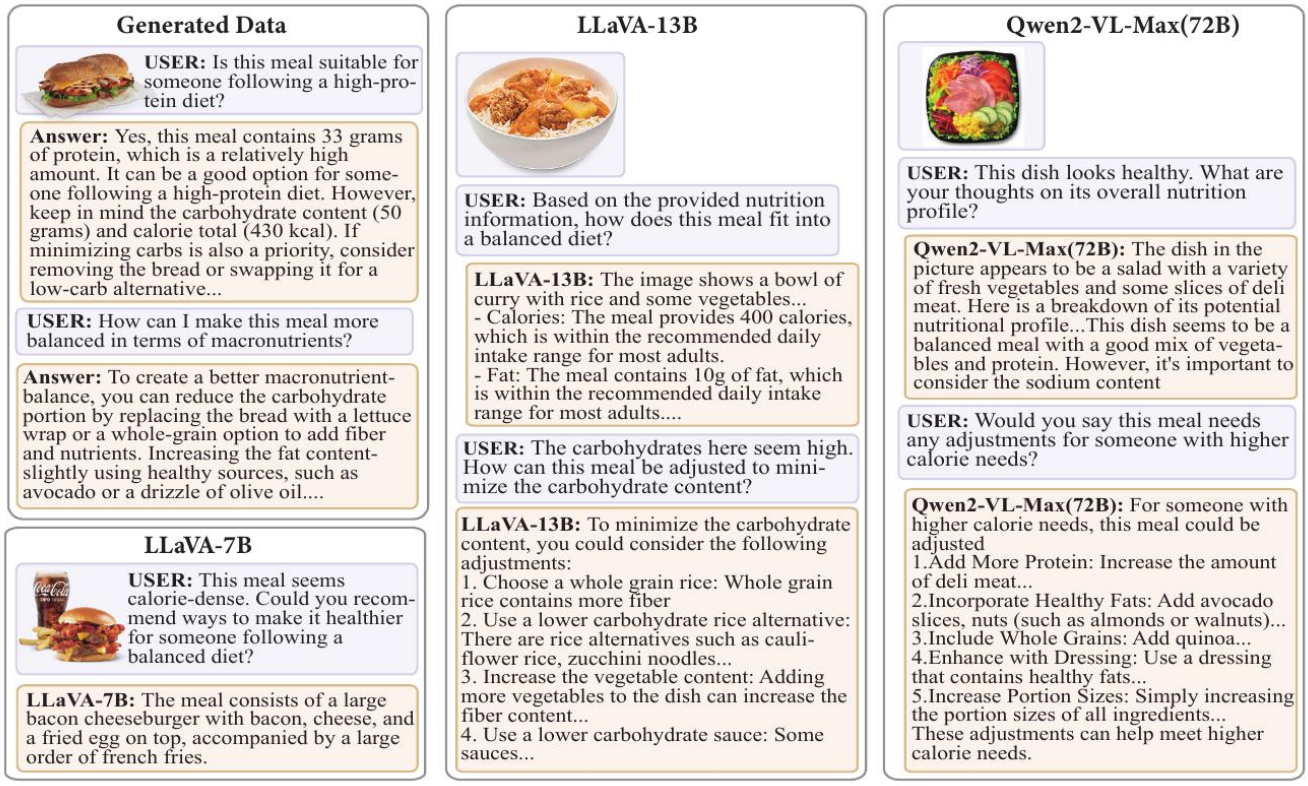

Figure 8: Examples from the auxiliary dialogue resource, including generated conversations and zero-shot LMM responses.

Nutrition5k, showing that ingredient semantics are a strong complement to visual features for monocular nutrition estimation. Our analyses further show that intermediate-layer fusion, robust ingredient aggregation, and majority-voted LMM ingredient prediction all matter in practice. Together, the benchmark and method provide both a stronger baseline and a useful resource for future research on ingredient-aware food understanding.

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